

Minseong Cho

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🏢 Softlab, Ajou University
📍 Suwon, South Korea
📁 Portfolio 
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Research Interests

Active matter dynamics · Elastic Leidenfrost effect · Mechanics of Moisture-Driven Hair Frizz

Education

- **M.S. in Mechanical Engineering** Sep 2025 – Present
Ajou University, Suwon, South Korea
 - **Advisor:** Prof. Jonghyun Ha
- **B.S. in Mechanical Engineering** Mar 2020 – Aug 2025
Ajou University, Suwon, South Korea

Research Experience

- **Graduate Research Student** – Softlab, Ajou University Sep 2025 – Present
 - **Mechanics of Moisture-Driven Hair Frizz**
 - Focus:** Moisture-driven changes in strand morphology.
 - Techniques:** Centerline-based geometric analysis and simulation-assisted interpretation.
 - Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung
 - **Boundary-Driven Active Matter in Soft Hydrogels**
 - Focus:** Leidenfrost-induced dynamics in soft hydrogel systems.
 - Techniques:** Image-based tracking, curvature analysis, and visualization scripts.
 - Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung
- **Undergraduate Research Student** – Softlab, Ajou University Mar 2023 – Aug 2025
 - **Boundary-Driven Active Matter in Soft Hydrogels**
 - Focus:** Leidenfrost-related soft-matter dynamics.
 - Techniques:** Image-based tracking, quantitative analysis, and figure preparation.
 - Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung

Conferences and Presentations

- APS Division of Fluid Dynamics Meeting** (Houston, TX, USA) Nov 2025
 - Leidenfrost-induced active matter dynamics of spherical hydrogels
- Asian Congress of Fluid Mechanics** (Seoul, South Korea) Sep 2025
 - Leidenfrost-induced active matter dynamics of spherical hydrogels
- KSME Annual Conference** (Jeju, South Korea) Nov 2024
 - Explosive motion of spherical hydrogel via the Leidenfrost effect

Awards & Honors

- **Dean's List** – Ajou University Spring 2024, Spring 2025
Recognized for outstanding academic performance (top 5% of the department).
- **Capstone Design Competition** – Ajou University June 2025
2nd Place · KRW 1,000,000 prize
Project: Compliant-mechanism-based knee assistive device for degenerative osteoarthritis prevention.
- **Gyeonggi-do Assistive Device Idea Competition** Nov 2025
3rd Place · KRW 300,000 prize
Organized by: Gyeonggi Province Rehabilitation Engineering Service Support Center (South Korea)
Project: Compliant-mechanism-based knee assistive device for older adults.
- **APS DFD Travel/Enabling Grant** 2025
US\$750 travel cost reimbursement
Organized by: APS Division of Fluid Dynamics, American Physical Society
Purpose: Travel support for participation in the APS Division of Fluid Dynamics Meeting.
- **Gallery of Fluid Motion** Apr 2026
Haecheon Choi GFM Award · KRW 400,000 prize
Organized by: The Korean Society of Mechanical Engineers, Fluid Engineering Division
Title: Explosive motion of spherical hydrogel via the Leidenfrost effect
Authors: Minseong Cho, Yeonsu Jung, Jonghyun Ha

Publications

- **Boundary-driven active matter and stochastic rectification via Leidenfrost impulses**
Minseong Cho, Yeonsu Jung, and Jonghyun Ha
✉ In preparation
- **Poroelastic swelling time as a quantitative marker of hair damage and treatment**
Jaehyuk Cho, Minseong Cho, Selin Cho, Haesoo Park, Hyun-Na Kim, and Jonghyun Ha
✉ In preparation

References

Prof. Jonghyun Ha

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Dr. Yeonsu Jung

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